a Source provenance

Public accession + processed file

Hash-verified bytes
Sample IDs and design
Transformations and feature IDs

b Frozen analysis contract

Settings + algorithm digest

Reference construction
Backend and random seeds
Declared resampling unit

c Numerical evidence

Per-cloud and per-study records

Bar lengths and reference draws
Candidate and reduced-cloud counts
Effect sizes and abstention

d Regeneration and scrutiny

Verification and regeneration

Inspect committed evidence
Download and regenerate analyses
Editable publication figures

Traceability supports scrutiny; it does not imply biological certification.